

Grace Bianchi

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EDUCATION

University of California, Santa Barbara

B.S. Statistics and Data Science, B.A. Economics

Expected Graduation: June 2027

3.88/4.00 GPA

Honors Thesis in Economics, Data Science Capstone – Admitted, Fall 2026

Relevant Coursework: Stochastic Processes, Probability Theory, Game Theory, Linear Algebra, Mathematical Statistics, Statistical Machine Learning, Regression Analysis, Econometrics, Vector Calculus, Differential Equations, Discrete Math

PROFESSIONAL EXPERIENCE

MUFG | Data Quality and Solutions Summer Analyst

June 2026 – August 2026

- Extracted 10 years of loss-run data from PDFs using Python, then cleaned and structured claims data in PostgreSQL
- Modeled claim frequency with Poisson and severity with Pareto distributions, using Monte Carlo simulation to estimate umbrella layer exceedance probabilities and support a coverage reduction, saving \$250K in premiums
- Validated reverse-engineered financial crimes rules in SQL during a platform migration for legacy-to-new parity

SageView Advisory Group | Data and Analytics Intern

June 2025 – August 2025

- Built a client segmentation model in R using weighted z-scores to score and tier 150+ institutional clients, and applied a decision tree to generate targeted improvement recommendations based on lowest-performing metrics
- Developed a SQL and PowerBI dashboard to search and visualize nested account, plan, and contact data
- Prepared investment reports for client meetings, compiling portfolio holdings and YTD returns

University of California, Irvine Spatial Neuroscience Lab | Research Assistant

June 2022 – January 2023

- Developed Python scripts to process VR-based spatial navigation experimental data for statistical analysis
- Supported integrating VR, fMRI, and diffusion-weighted imaging to study neural systems underlying navigation
- Assisted in EDA and scientific literature review on hippocampal and cortical contributions to spatial cognition

PROJECTS

Markov Regime-Switching Volatility Forecaster

- Built a Python volatility forecaster combining GARCH, HAR-RV, and a BIC-selected 4-state HMM, validated across five major SPY market events including COVID and the 2025 tariff selloff
- Developed a regime-aware VaR framework using Markov-switching forecasts and Extreme Value Theory, backtested over 887 days with Kupiec and Christoffersen coverage tests
- Simulated regime-dependent volatility distributions and used Diebold-Mariano tests to improve tail calibration against HAR-RV

Stellar Classification: Using Machine Learning to Assess the Role of Redshift

- Fit 14 classification models (7 types x full/reduced feature sets) in R to classify stellar objects from SDSS data
- Demonstrated that redshift is a critical predictor – removing it drops Random Forest test accuracy from 97.9% to 84.1%, with STAR classification most impacted (39% misclassification rate without redshift)
- Tuned hyperparameters via 10-fold cross-validation; evaluated best models using ROC AUC, accuracy, ROC curves, and confusion matrices

Housing Market Segmentation using Clustering and Association Rules

- Applied k-means, hierarchical (Ward.D2), and PAM clustering to 1,047 homes in R, selecting k=3 via elbow method and validating with Hopkins statistic (H=0.999) and cophenetic correlation
- Identified three market segments (entry-level, mid-market, high-value) with PCA confirming a dominant size-value axis explaining 49.1% of variance across 7 housing features
- Mined 6,749 Apriori association rules, finding that a large living area and 3+ bathrooms predict an expensive price tier with 100% confidence and a lift of 2.94

SKILLS & ACTIVITIES

Programming Languages: Python, R, SQL, Java, Stata

Tools: PostgreSQL, Git, Jupyter, Power BI, Excel, RStudio, LaTeX

Awards and Honors: UCSB Dean's Honors, AP Scholar with Distinction, AMC Math Competition Participant

Activities: Undergraduate Learning Assistant: Intermediate Microeconomic Theory (Summer 2026), Statistics for Economics (Fall 2026 - Spring 2027), UCSB Data Science Club (2024-2026), UCSB Data Science Collaborative (2024-2026), Kappa Alpha Theta (2023-2025), Scitrek (2024)

Interests: Sudoku (Extreme PR: 4 minutes 56 seconds), WWII history, Risk, Fleetwood Mac